Slope and Equations of Lines

Concept

Slope measures rate of change. A line can be written in several equivalent forms.

Slope

$m = (y_2 - y_1)/(x_2 - x_1)$, for different x-values.

Forms

$y = mx + b$; $y - y_1 = m(x - x_1)$; $Ax + By + C = 0$.

Parallel / perpendicular

Equal slopes / slope product -1, when defined.

Special lines

Horizontal: $y = b$, slope 0. Vertical: $x = a$, slope undefined.

Example 1

Find the line through (2,3) and (6,11).

Solution

1. $m = (11 - 3)/(6 - 2) = 2$.

2. $y - 3 = 2(x - 2) \rightarrow y = 2x - 1$.

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Example 2

Find the line through $(2,5)$ perpendicular to $y = (2/3)x - 1$.

Solution

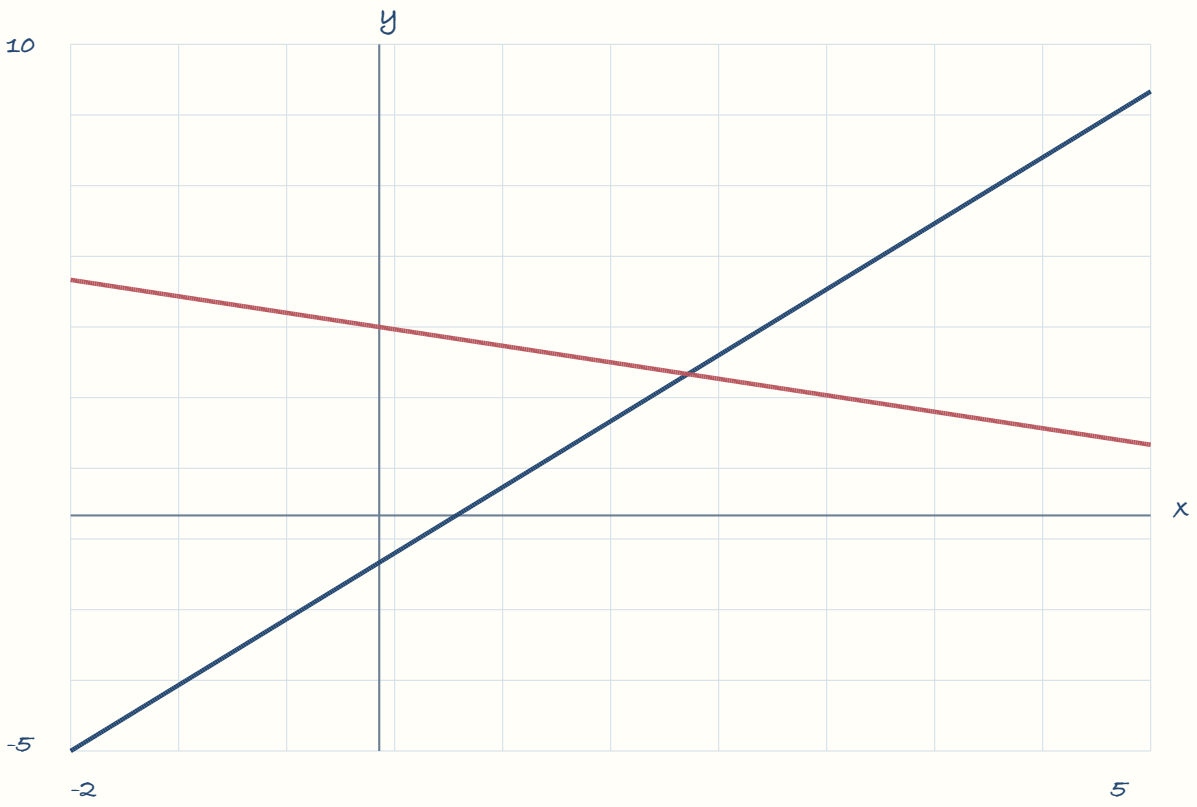
1. Perpendicular slope $= -3/2$.

2. $y - 5 = (-3/2)(x - 2) \rightarrow y = (-3/2)x + 8$.

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Visual notes | read the graph

Blue: $y = 2x - 1$; red: $y = -x/2 + 4$



The slopes multiply to −1. These lines are perpendicular, even if unequal axis scales alter the visual angle.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find slope through (-1,4) and (3,-8).

2. Write a line through (2,7) with slope 3.

3. Find the line perpendicular to $y = -x/4 + 2$ through the origin.

4. Write the vertical line through (5,-2).

5. Are $y = 2x + 1$ and $4x - 2y + 2 = 0$ distinct parallel lines?

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Practice answers | check after trying

1. $(-8 - 4)/(3 + 1) = -3$.

2. $y - 7 = 3(x - 2)$, so $y = 3x + 1$.

3. $y = 4x$.

4. $x = 5$.

5. No. The second simplifies to $y = 2x + 1$; they are the same line, with infinitely many common points.